

BEGINNER

pedro



STEM Program

Lesson #10: Becoming a Pedro Maker



Complete this lesson
to earn your Pedro
Badge

Add it to your Pedro
Passport to track
your STEM journey

YOUR MISSION: Explore every Pedro mode in this lesson and
become a true Pedro Maker!

Learning Goals

In this lesson, you will explore all of Pedro's operating modes and unlock the full potential of the Pedro robot. By the end of the lesson, you will be able to configure Pedro in each mode, understand how each communication method works, and confidently use the different communication interfaces.



At the end of this lesson, you will unlock your Pedro Badge.

 the Pedro Maker Badge

Teacher Notes

- No prior robotics experience required.
- Duration: 60–90 minutes
- Difficulty: ● Beginner
- Required Materials
 - **2 Assembled** Pedro robot **per student or per group** (*recommended: groups of 2 students*)
 - **OLED Screen 128X64**
 - **USB cable**
 - **NRF24L01 module**
 - **HC-05 module**
 - **2 jumper wires**
 - **1 computer** with the Arduino IDE installed



Think: The **brain icon** encourages students to make careful observations, share their ideas, and engage in discussion by explaining and justifying their answers.



Learn: The **light bulb icon** indicates important concepts that students should read carefully. Students are encouraged to ask questions and discuss these topics about the robotics concepts presented in the lesson.



Make: The **pencil icon** indicates that students must complete the activity on the page. These activities are designed to reinforce the concepts learned through experimentation, and problem-solving.



Welcome to Your Final Challenge

Congratulations!

You have completed the Pedro STEM Program.

Throughout this journey, you discovered that robotics is much more than moving motors. A robot combines **mechanics, electronics, programming, communication, and engineering thinking** to perform useful tasks.

You built Pedro step by step, explored different communication technologies, learned how hardware and software work together, and programmed the robot.

Today marks the end of your Pedro lessons but it is also the beginning of your journey as a robotics maker.



Final Reflection

Answer the following questions.

Question 1

Why is a robot more than a simple machine?

Question 2

Which Pedro communication mode did you find the most useful?

☐ Manual ☐ Radio ☐ Bluetooth ☐ UART ☐ USB ☐ REPLAY & RECORD

Explain your choice.

Question 3

What is the advantage of wireless communication?

Question 4

What is the advantage of wired communication?

Question 5

Which lesson was the most difficult? Why?

Question 6

Which lesson was your favorite? Explain.

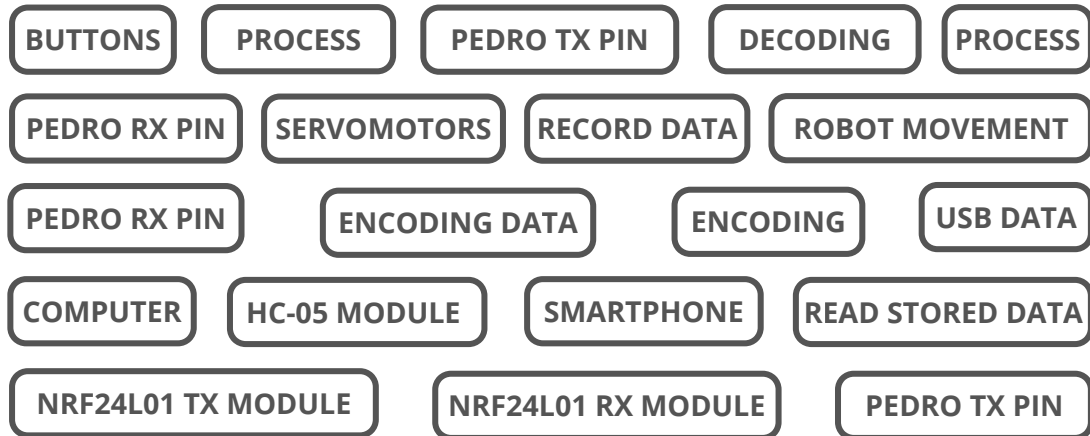
Question 6

Which lesson was your favorite? Explain.

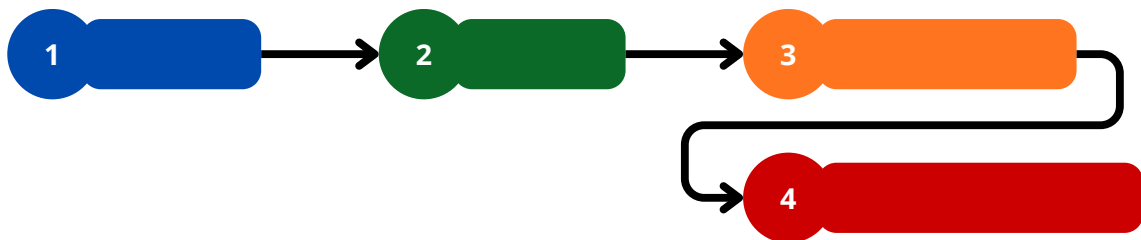


Final Challenge

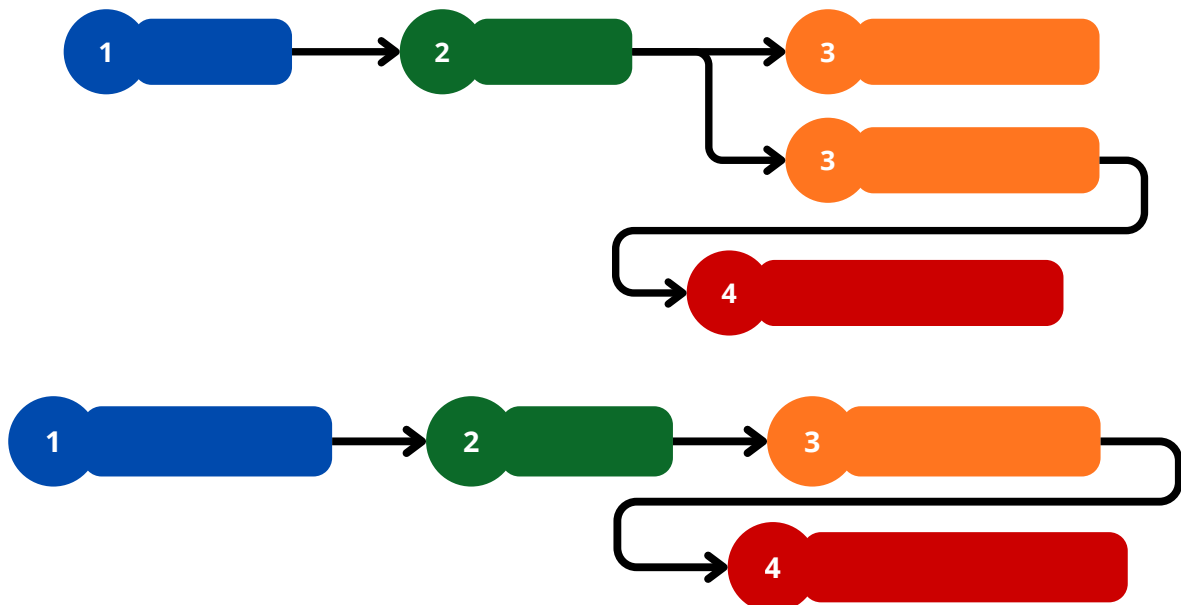
Without looking back at the previous lessons, complete the communication chain for each Pedro mode by filling in the missing steps.



Manual Mode Chain:



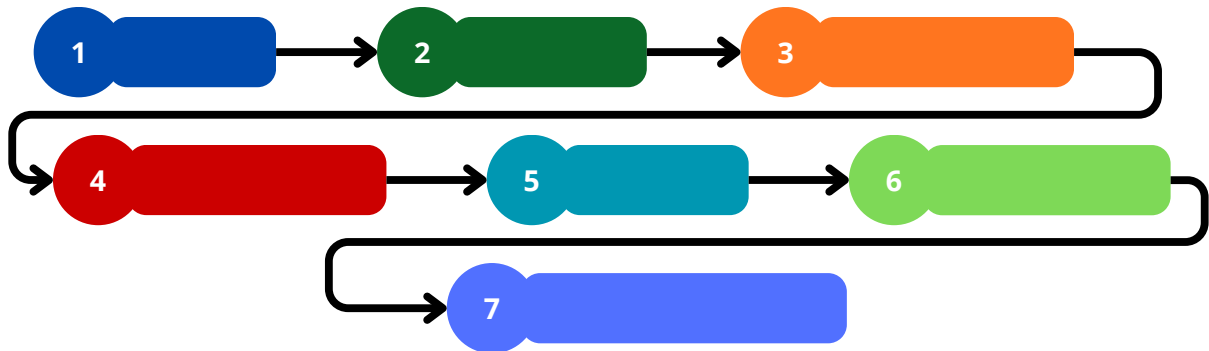
Record & Replay Mode Chain:



USB Serial Communication Chain:



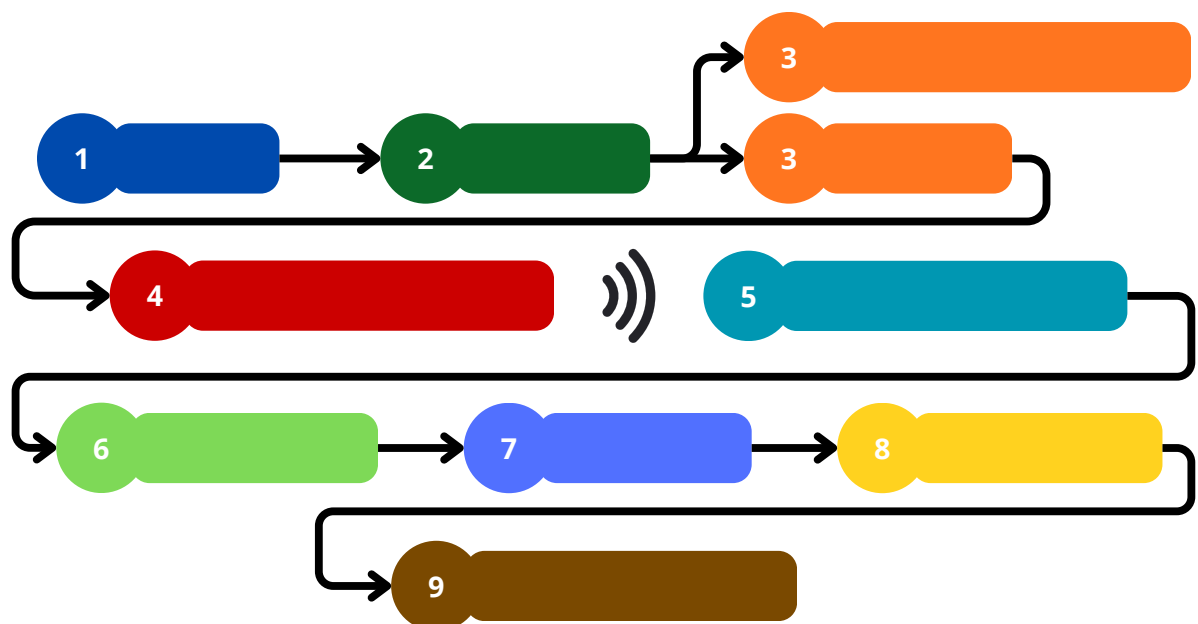
UART Communication Chain:



Bluetooth Communication Chain:



Radio Communication Chain:





Crossing Communication Modes

In this section, you will discover how to **combine multiple Pedro communication modes** by integrating **UART communication** with the other control modes.

This feature allows several Pedro robots to work together as a team. A Pedro robot can **receive commands wirelessly** (Radio, Bluetooth) and **forward the same commands** to other Pedro robots through the **TX/RX UART connection**.

This makes it possible to control multiple robots simultaneously from a single Pedro robot. To enable UART communication with the selected control mode, set the UART parameters to **TX or RX**.

The first Pedro robot that receives the user's commands is called the **Leader**. All other connected Pedro robots are called **Followers**. The Leader receives the commands and automatically distributes them to the Followers through the UART connection.

 **Note** : The **Follower** robots cannot be controlled using their own buttons. They only execute the commands received from the **Leader**.

Combination 1:

Leader	Follower #1
USB Mode + TX UART	UART Mode RX

Combination 2:

Leader	Follower #1
Bluetooth Mode TX + TX UART	UART Mode RX



Combination 1: USB Mode + UART

The computer sends digital commands to the Leader Pedro via the USB cable. The Leader processes these commands and forwards them through the UART connection, allowing the Follower robot to perform the same movements simultaneously.

Pedro Robots Configuration

Leader

1. Switch to **USB** mode.
2. On USB mode settings screen set **UART** to **TX**.
3. **OK** to validate USB mode settings.
4. Connect Pedro to the computer with a USB cable.
5. Verify that the **TX** indicator is displaying on the screen.

Follower

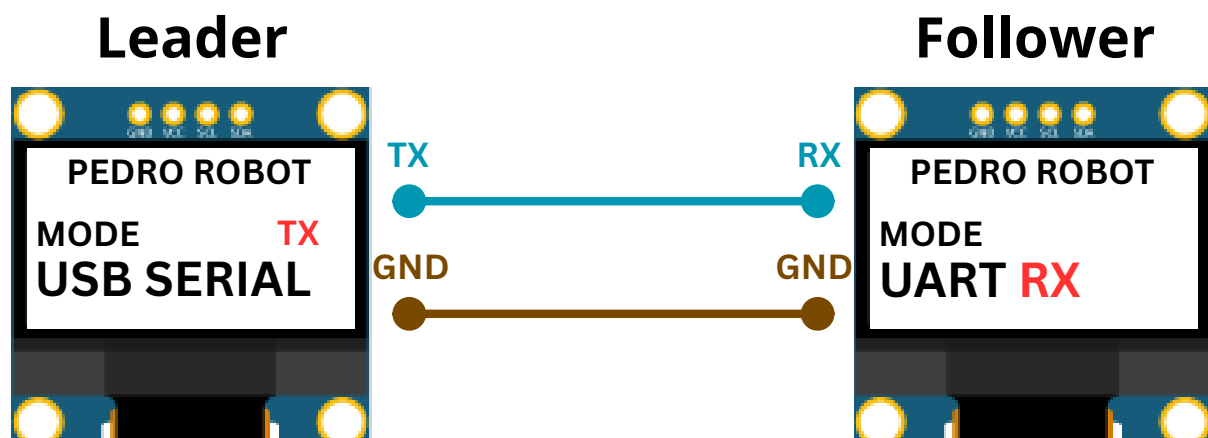
1. Enter **UART RX** mode.

 **UART Wiring:** Connect the Pedro robots pin as follows

1. Leader **TX** → Follower **RX**
2. Leader **GND** → Follower **GND**

Test the System

Open the Web Controller and connect it to the Leader. Then, use the control buttons in the Web Controller to operate the system. You will see all the Follower robots automatically reproduce the Leader's movements in real time.





Combination 2: Bluetooth Mode + UART

The Leader receives commands by Bluetooth then processes these commands and forwards them through the UART connection, allowing the Follower robot to perform the same movements simultaneously.

Pedro Robots Configuration

Leader

1. Switch to **Bluetooth Mode**.
2. On Bluetooth settings screen set **UART** to **TX**.
3. **OK** to validate Bluetooth settings.
4. Verify that the **TX** indicator is displaying on the screen.

Follower

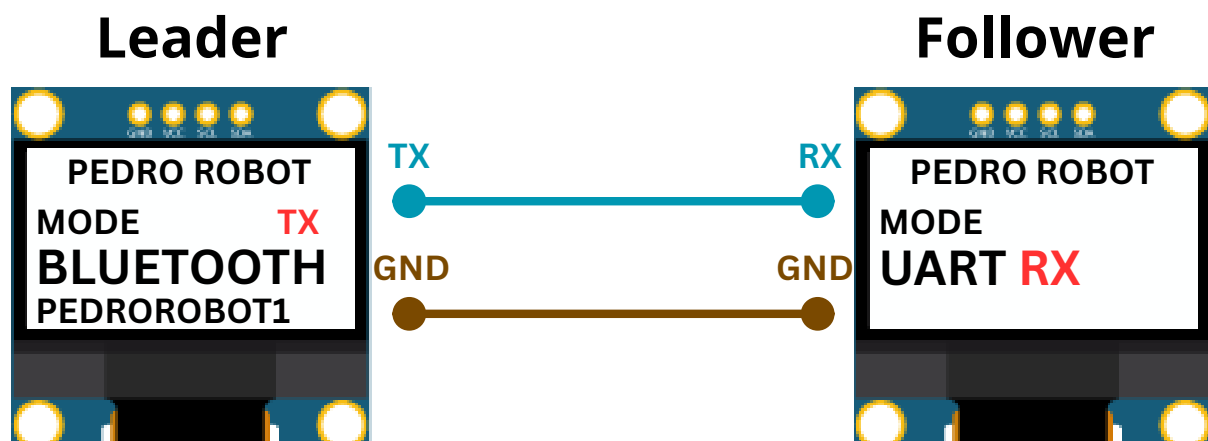
1. Enter **UART RX** mode.

 **UART Wiring:** Connect the Pedro robots pin as follows

1. Leader **TX** → Follower **RX**
2. Leader **GND** → Follower **GND**

Test the System

You can now pairing the Leader with your smartphone and connect it by bluetooth. You will see the Follower robot automatically reproduce the Leader's movements in real time when you send command from your smartphone.





Combination Challenge

Using everything you have learned throughout the Pedro STEM Program, design your own **multi-robot communication system**.

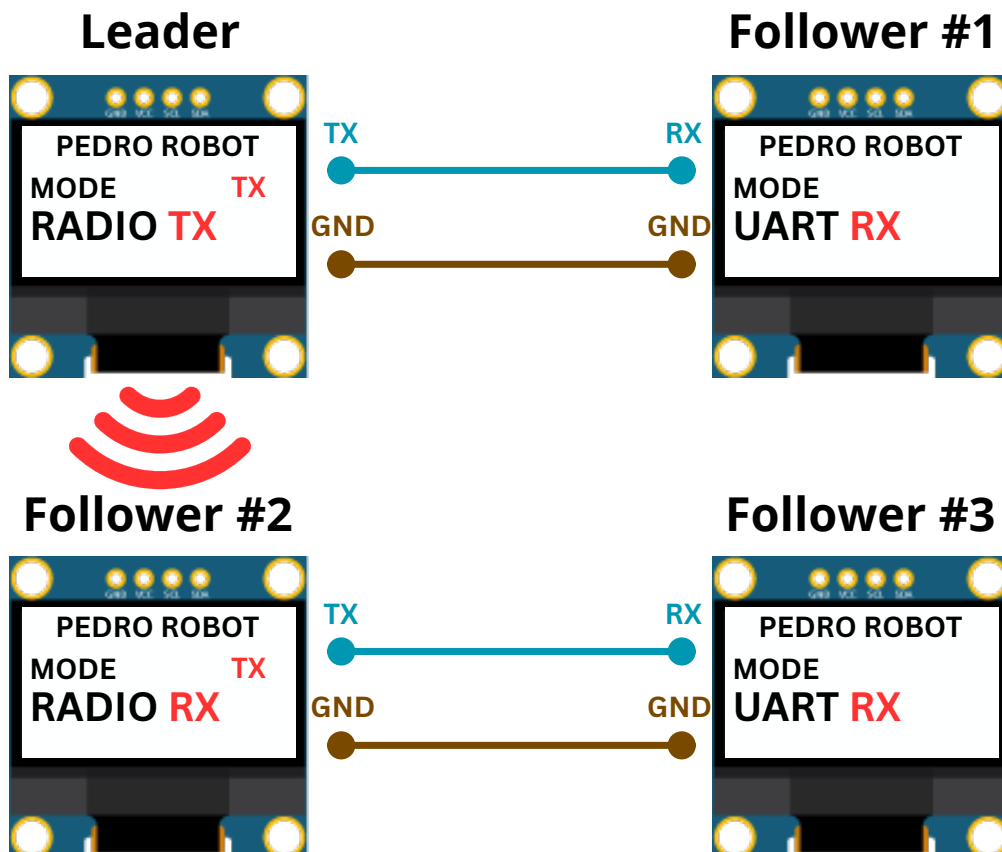
Configure the following Pedro robots:

- **Leader:** Radio Mode TX + UART TX
- **Follower #1:** UART Mode RX
- **Follower #2:** Radio Mode RX + UART TX
- **Follower #3:** UART Mode RX

For each Pedro robot:

- Configure the correct **operating mode**.
- Select the appropriate **robot role** (Transmitter or Receiver).
- Enable the required **UART bridge**.
- Complete the **UART wiring** between all robots.

Once the system is fully configured, control the **Leader** and observe how all the **Follower** robots automatically reproduce the Leader's movements in real time.





Final Quiz

1. **What is the role of a microcontroller?**
2. **Which component moves Pedro's arm?**
3. **What is PWM used for?**
4. **What does a servo motor do?**
5. **What is the difference between Manual Mode and Record Mode?**
6. **Which communication mode uses the NRF24L01 module?**
7. **Which communication mode uses the HC-05 module?**
8. **Which pins are used for UART communication?**
9. **What is the purpose of the Code Key in Radio Mode?**
10. **Why must TX connect to RX?**
11. **Why should two Pedro robots share the same GND?**
12. **What information is stored during Record Mode?**
13. **Why doesn't Pedro use inverse kinematics?**
14. **What is the purpose of the Arduino loop() function?**
15. **Which communication methods are wired?**
16. **Which communication methods are wireless?**
17. **Give one real-world application of wireless robotics.**
18. **Which engineering skill do you think you improved the most?**
19. **Imagine Pedro 2.0. What new feature would you add, and why?**

Congratulations Mission Accomplished!

You have reached the finish line! Throughout this STEM journey, you have discovered how a robot works by combining **mechanics, electronics, and programming**.

You learned how to:

-  Build your own robot using 3D-printed parts.
-  Understand the interaction between hardware and software.
-  Program Pedro and control its movements.
-  Explore the essential communication technologies used in robotics, including **Manual, Radio, Bluetooth, UART, and USB**.
-  Think like an engineer by solving problems through experimentation and observation.

By mastering all of Pedro's communication modes, you have taken your first steps into the exciting world of robotics and embedded systems.

This may be the end of the Pedro STEM Program, but it is only the beginning of your engineering journey.

Keep building, keep creating, keep experimenting, and never stop asking "**How does it work?**"

Today, you officially become a...



Pedro Robot Maker



pedro

STEM PASSPORT

Your Journey. Your Skills. Your Future.



This Passport Belongs To:

★ Collect all 10 badges and become a Pedro Maker! ★

MY PEDRO BADGES

Complete each lesson to earn your badge!

1 PEDRO EXPLORER Date: _____	2 PEDRO BOARD EXPERT Date: _____	3 PEDRO BUILDER Date: _____	4 PEDRO MOTION ENGINEER Date: _____	5 PEDRO RADIO SPECIALIST Date: _____
6 PEDRO AUTOMATION EXPERT Date: _____	7 PEDRO BLUETOOTH ENGINEER Date: _____	8 PEDRO SERIAL PROGRAMMER Date: _____	9 PEDRO ROBOT DESIGNER Date: _____	10 PEDRO MAKER Date: _____

CONGRATULATIONS!

You completed all 10 lessons and unlocked all Pedro badges.
You are now an official

★ **PEDRO MAKER!** ★